

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	FF-NIA19-0000000050
Award Program	2019 New Innovator in Food and Agriculture Research Award
Project Title	Exploring plant-microbiome-environment interactions towards tailoring plant disease management
Grant Start Date	01/01/2020
Grant End Date	04/01/2024
Total Grant Budget	\$581992.52
Total FFAR Budget	\$290893.65
Matching Funder(s)	Auburn University and University of Florida
Final Report Date	04/01/2024
Project Director/Principal Investigator Information	
Full Name	Neha Potnis
Email	nzp0024@auburn.edu
Phone	334-844-2524
Grantee Organization Information	
Organization Name	Auburn University
Address	209 Rouse Life Science Building
	Department of Entomology and Plant Pathology
City, State Zip	Auburn, AL 36849



Tax ID	63-6000724
Authorized Signing Official (ASO) Information	
Name	Anthony Ventimiglia
ASO Title	Assistant VP for Research Administration for SVPRED
Email	ventiaf@auburn.edu
Phone	334-844-5954

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

[Auburn, Alabama](#)
[Gainesville, Florida](#)

- 1.2. How many new jobs were created because of this grant funding?

[9 graduate students, 1 research associate, and 1 post-doctoral researcher mentored in this project.](#)

- 1.3. How many jobs were maintained because of this grant funding?

[9 graduate students, 1 research associate, and 1 post-doctoral researcher supported because this grant funding.](#)

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

[No.](#)

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

[No.](#)

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of



the project help to address important needs in U.S. food and agriculture systems.

Traditional disease management strategies for endemic diseases are often limited in their success of preventing recurring outbreaks. Growers continue to bear losses due to these outbreaks that limit profitable crop production. This project addressed a major endemic disease of tomato and pepper, known as bacterial leaf spot (BLS), that is a recurring and yield-limiting issue to the growers around the world. This project sampled the tomato fields in the southeastern U.S. for the three consecutive years, assessed the variable disease pressure, traced the changes in pathogen population and evaluated the risk factors such as climatic parameters and production practices for their contribution to continued disease outbreaks. A significant finding from this work was that climatic fluctuations and extremes are major drivers for fueling disease outbreaks in the southeastern US. One of the crucial steps in controlling the BLS is to limit pathogen on the transplants that get planted in the field. Thus, the second goal of this project was to devise microbiome-informed seed treatments that can help limit pathogen transmission from the seed-to-seedling stage. This project has established a large collection of tomato-associated microbes and screened this collection as an option for the seed treatment. We have identified variable contribution of different species of microbes in limiting pathogen on seedlings. This project has made a significant contribution to the understanding of tomato-associated microbiome and a step towards developing microbiome-informed seed treatment. The findings from this work are expected to provide timely solutions to the current production problems that are limiting farmers' profitability.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

Following are the key findings from this project that can be used by the agricultural community:

1. Disease prediction models: Field survey of tomato coupled with rich climatic, and field associated metadata collection and mathematical and statistical modeling approaches have allowed development of disease prediction models in this project. These models will help in disease surveillance under global changes. These findings are currently being written in the manuscript and models will be made available as a part of this manuscript. We think that these models will not only be useful for tomato endemic diseases but also for other endemic diseases in the southeastern U.S.
2. Synthetic microbial communities for further exploration as therapeutics: We have identified core microbial members of the community that contribute towards



reduced BLS disease severity when inoculated onto the seeds. This concept of microbiome-informed engineering was developed in this project and tested for its efficacy. This approach can serve as a viable solution to tackle management of this recurring endemic pathogen.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Potential outcomes that could impact U.S. agriculture and food systems:

1. **Development of high-resolution approach for pathogen surveillance:** Understanding the pathogen diversity and drivers of this diversity is key to devising disease management strategies. This project developed methodology to study pathogen population structure using cultivation-independent approach. This approach allowed overcoming the inherent bias in culture-dependent methods of focusing only on dominant pathogen population. The culture-independent method i.e. shotgun metagenome sequencing of field samples allowed tapping into pathogen variants and tracing these variants over the course of the growing season. We published pilot study using this approach in Environmental microbiology (Newberry et al. 2020). One of the major findings from this work is that co-occurrence of multiple pathogen genotypes is very common across the fields. This changed our view of binary plant-pathogen interactions to adopting multiple pathogen genotypes-host framework. We also found that higher pathogen diversity correlated with higher disease severity. This finding meant that introduction of multiple pathogen genotypes via multiple inoculum sources may be a factor in fostering recurring outbreaks. Our modeling approaches indicated that climatic shifts have been majorly responsible for driving disease dynamics of BLS endemic disease. These findings and methodologies will be largely applicable to understanding drivers of dynamics of other endemic diseases. The concern of climatic shifts in fueling disease outbreaks means that we need to understand the influence of climatic fluctuations not only on pathogen or host alone, but also through a pathogen-



host-microbiome interaction framework, since microbial communities have been hypothesized to offer stress resilience.

2. **Moving beyond binary interactions between single pathogen and single host:** Traditionally, disease management strategies have been designed with evaluation of a single pathogen on host. However, in nature, it is rare to find a single pathogen and often multiple co-occurring pathogens, or opportunistic pathogens are common. We conducted greenhouse experiments to see whether multiple co-infecting pathogens belonging to different species or genera may have influence of the overall disease outcome. Indeed, the interaction partners determined the outcome of the disease. We also found that coinfection does not always lead to increased disease severity. Another aspect of coinfection experiments involved inoculation of multiple pathogen genotypes or lineages on a single host. We found that certain pathogen genotypes lead to higher disease severity and certain combinations lead to lower severity, indicating that these interactions might be direct strain-to-strain interactions or strain-host interactions mediating dynamics of multiple pathogen lineages. We find that humidity plays a crucial role in driving coinfection dynamics. We discuss these complex plant-microbe-environment interactions and conceptual advances in the Annual Review of Phytopathology article (Potnis, 2021).
3. **Microbial Consortia for mitigating disease:** A major outcome of this project is a collection of microbial isolates from tomato seeds, leaves, from endosphere, rhizosphere, and phyllosphere compartments. This collection will be an asset for a variety of future experiments by our group as well as other scientific community interested in exploring this collection. This collection and parallel amplicon sequencing data allowed selection of core microbiota members for synthetic community experiments. These synthetic microbial consortia have been screened for their ability to limit disease on seedlings. The future work can be performed on identifying effective methods to incorporate these community members as inoculants on the seeds. This work has a potential to lead to development of new product for managing endemic seed-borne disease.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Goal 1: Identify the risks factors associated with BLS disease severity, yield losses, and pathogen diversity

Obj 1.1. Survey of tomato fields throughout southeastern US, metadata collection, sample collection

Obj. 1.2 Assess the genetic landscape of the BLS pathogen



Obj 2.1 Identify risk factors that lead to higher BLS severity and yield losses

Obj. 2.2 Assess variable importance of risk factors

Goal 2. Leverage the knowledge of pathogen diversity and interaction with microbiome towards tailoring disease management strategies

Obj. 3.1 Identifying relationship between pathogen diversity and disease severity- Greenhouse experiment

Obj. 3.2 Effect of host genotype and associated microbiome on bacterial spot disease severity- taxonomic profiling of metagenome data, network analysis

Obj. 3.3 Identify a cocktail of endophytes for seed treatment

Obj 4. Modeling socio-economic impact of BLS management strategies

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

We had a limited success in collecting all field metadata from the survey of tomato fields in the southeastern U. S. There was less participation in survey questions that asked for specific usage of pesticides or fertilizers, tomato genotypes and the estimate of yield. Thus, these parameters could not be incorporated in the analysis of variable contributions of these factors towards disease severity and incidence. However, other parameters, particularly environmental parameters and production scale provided a rich dataset to evaluate the risk factors.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Goal 1: Identify the risks factors associated with BLS disease severity, yield losses, and pathogen diversity

Obj 1.1. Survey of tomato fields throughout southeastern US, metadata collection,



sample collection

Field surveys in the tomato fields in Southeastern US have been conducted for the past three years and data compiled for these three years as proposed in the project. A total of 30 fields have been sampled for mid-and end season over three consecutive years. The sampled fields included both small scale and commercial. Along with the samples, the extension agents from each state helped us to collect metadata such as disease severity, pattern of pesticide and fertilizer use, history of bacterial leaf spot in the field, and their socio-economic status. The metadata was compiled and analyzed for identifying climatic factors as well as other agronomic practices as risk factors for disease severity.

Shotgun metagenome sequencing was conducted on all the phyllosphere tomato samples. Along with cultivation-independent approach, cultivation-based approach has also been followed. Over the past three years, we have obtained a collection of *Xanthomonas* isolates from various fields, which represents a large spatio-temporal pathogen collection.

Obj. 1.2 Assess the genetic landscape of the BLS pathogen

A total of 66 field samples were obtained over the period of three years, 2020-2022. *Xanthomonas perforans* was the only dominant species in tomato fields, although the disease severity varied across the geographical locations. Opportunistic pathogens such as *Xanthomonas arboricola* and *Pseudomonas cichorii* were present co-occurring with bacterial spot *Xanthomonas* sp. Further resolution into infrasubspecific level for bacterial spot *Xanthomonas* using analysis of high-resolution shotgun metagenomics data indicated that each state showed distinct pathogen lineage dominating, although multiple pathogen lineages (up to five) were present co-occurring in each field and the pathogen dynamics varied across fields within each state, indicating size/scale of production and agricultural practices may be driving forces for such variation. Interestingly, this three-year survey also indicated that all pathogenic lineages of *X. perforans*, that are found globally (Sequence clusters 1 through 8) are all circulating in the southeastern US tomato fields.

Obj 2.1 Identify risk factors that lead to higher BLS severity and yield losses

We collected field metadata that included size or scale of field operation (small vs commercial) and the climatic factors. This metadata was used for disease modeling approach. We used disease severity as our response variable and considered climatic factors, sampling time, year, and farm-scale as predictors, treating the latter three as categorical variables. In addition to routine climatic variables, their mean or median values, we also considered upscaling approach, that allows considering the extreme climatic values. We employed an ordinal logistic regression model to analyze the relationship between their predictors and ordinal disease severity. This approach helped identify key factors influencing the severity of BLS disease. Although not significant, the size of the sampling field positively influenced the disease severity, suggesting that an increased sampling field size or scale (commercial scale) would increase the possibility of increased disease severity. A



similar observation was made about sampling year, suggesting that in 2020, the BLS disease severity in the sampled field was higher than in the years 2021 and 2022. Among the climatic variables, the predictor average of clear sky surface photosynthetically active radiation (PAR), indicative of the photosynthetic capability of plants and changes in the average values of the wind direction at 10 meters above the surface of the earth had the significant positive influence on BLS disease severity. The frequent shifts in wind direction and extreme changes in surface pressure and relative humidity influenced *Xp* abundance. These findings indicate that climatic shifts are major contributors for recurring disease outbreaks. This study provides valuable information for disease diagnostics, epidemiological studies, and predictive modeling, emphasizing the importance of high-resolution metagenomics and considering meteorological parameter distribution in disease management and early warning systems.

Obj. 2.2 Assess variable importance of risk factors influencing pathogen diversity
Correlation analysis between the pathogen diversity and disease severity suggests that the diversity of *Xp* strains positively correlated with disease severity, suggesting that multiple lineages may increase the pathogen's ability to cause disease. Changes in strain composition were evident across different fields and within individual fields throughout the growing season, spanning three years. These dynamics in strain composition were marked by fluctuations in their relative abundance, with a notable prevalence of introducing new lineages during the end season, introducing entirely new strains, and the dominance of one strain over others. Next, we evaluated influence of climatic factors in explaining the differential dynamics of pathogen lineages across different fields. We found that the variables that appeared to be significant across clusters were shifts in relative humidity and time of sampling, particularly mid-season, that appeared to diminish abundance across all clusters. Also, we identified specific environmental variables that may have negative impact on abundance of particular lineages, suggesting that apart from strain-strain interactions, climatic variables may also influence strain dynamics across neighboring fields. The findings from objective 1 and 2 have been written as a part of two student theses and, also as a manuscript. We are currently working on finalizing the draft of this manuscript and hope to communicate with the peer-reviewed journal by the end of April.

Goal 2. Leverage the knowledge of pathogen diversity and interaction with microbiome towards tailoring disease management strategies

Obj. 3.1 Identifying relationship between pathogen diversity and disease severity-
Greenhouse experiment

One striking observation from the strain-level resolution shotgun metagenome data is that multiple sequence clusters of *X. perforans*, a dominant pathogen in the tomato fields in the southeastern US, co-occur in individual fields and their relative abundance levels vary with seasonal patterns, indicative of seasonal



pathogen population dynamics. In order to dissect interactions among different sequence clusters and influence of coinfection on disease severity, we have conducted a greenhouse experiment with permutations/combinations of strains belonging to different sequence clusters carrying specific fluorescent and antibiotic markers (that allow tracking of each sequence cluster in planta over time) were inoculated onto susceptible tomato cultivar and subsequently tracked using culture-dependent (plating on selective antibiotic media) and culture-independent approach. The results indicate that while the strains belonging to different sequence clusters do not exhibit any antagonism against each other under in vitro conditions, they do exhibit inter-strain interactions under in planta conditions. Individual strains vary in their degree of disease severity on susceptible cultivars under greenhouse conditions and the combinations with more distantly related strains results in higher disease severity. Humidity had a significant influence on overall in planta *Xanthomonas* population, as well as on disease severity in presence of mixed infection. Coinfection with more than two genotypes resulted in lower disease severity under high humidity conditions, however, such difference in disease severity was not evident under low humidity conditions. Next, we conducted additional experiments that traced coinfection dynamics of two strains (a strain belonging to most abundant lineage and the strain belonging to the least abundant lineage under field conditions). We observed that while proportion of inoculum of each strain in coinfection may influence the population dynamics, the strain belonging to field-dominant lineage also dominated in the greenhouse experiments, indicating its higher in planta fitness likely is responsible for its dominance in the fields. The coinfection dynamics experiments provided opportunity to explore the technique of flow cytometry for accurate quantification of pathogenic strains during coinfection, avoiding the bias that may be inherent to cultivation-dependent methods.

Obj. 3.2 Effect of host genotype and associated microbiome on bacterial spot disease severity

In this objective, we evaluated whether nematode-resistant tomato genotypes, routinely deployed in southeastern U.S. show predisposed susceptibility towards BLS pathogen. Nematode resistant tomato lines were screened for their susceptibility towards bacterial spot pathogen in presence and in absence of nematodes (three different species, *javanica*, *hapla*, and *incognita*). Eggs per gram of root were counted over the course of 90-day experiment. We did not find any influence of nematode inoculation on bacterial spot disease severity and there was no statistically significant difference in *Xanthomonas* population on nematode resistant vs susceptible plants under greenhouse conditions.

Since the field survey also indicated that presence of opportunistic co-occurring pathogens was common in *Xanthomonas*-infected plants, we conducted in planta mixed infection experiments with *X. perforans* with the other co-occurring pathogen, *Pseudomonas capsici*. Dip-inoculation experiments under different humidity conditions have shown effect of humidity on overall disease severity as well as lower populations of *X. perforans* in presence of *Pseudomonas capsici*. Our results from fluorescence microscopy, nutritional overlap experiments as well as



indicate that *P. capsici* benefits from susceptibility induced by *X. perforans*. *P. capsici* induces plant innate immune response that reduces *X. perforans* populations during initial stages of the disease development. This work is now published in Plant Disease.

Obj. 3.3 Identify a cocktail of endophytes for seed treatment

Our objective here is to obtain seed-associated bacterial isolates that can be evaluated for their potential to prevent the transmission of disease caused by *Xanthomonas* from seed to seedling stage. A collection of fifty bacterial isolates comprising 4 bacterial phyla (*Firmicutes*, *Actinobacteria*, *Bacteroidetes*, and *Proteobacteria*) and 18 bacterial genera were obtained from tomato seeds and termed as a “healthy” microbiome under the assumption that they were isolated from seed otherwise healthy and free of disease. *Proteobacteria* made up the largest phylum of cultured isolates (46%), followed by *Firmicutes* (22%), *Bacteroidetes* and *Actinobacteria* (14% each), with 16% being unresolved due to either no matching results in NCBI blast or matching to only strains with no taxonomical characterization. These community members were further tested for their antagonistic interactions against *X. perforans*. *Bacillus*, *Pantoea* sp. were identified to show strong antagonistic activity against *X. perforans*. A synthetic community composed of isolates representative of each family was inoculated onto the seeds to track colonization of these community members onto rhizosphere, endophere or phyllosphere. We observed the differences in microbial community structure on *Xanthomonas*-infected plants when treated with the synthetic community compared to the plants treated with synthetic community alone. This experiment also allowed us to identify community members that show differential abundance upon *Xanthomonas* infection. Next, we identified isolates from our tomato microbial collection that matched to their 16S and/ ITS sequence to the core microbiome members. A cocktail of 12 strains in combination and individually were tested in its contribution towards limiting disease severity on tomato seedlings. In addition, individual strains and their contribution to overall plant health parameters such as root weight, pattern, shoot weight and length were investigated. These permutation and combination experiments were conducted in the growth chamber settings upon inoculation of seed and following the disease severity on 21-day old seedlings. We are conducting experiments to quantify *Xanthomonas* population with the select combination of synthetic community members. In parallel to this experimental approach, we used the data from the permutations to the machine learning approach to predict the phenotype of disease severity and plant growth for the rest of the synthetic community combinations. These predictions are guiding future experiments for proof-of-concept study. We are in the process of writing the manuscript draft for this work.

Obj 4. Modeling socio-economic impact of BLS management strategies

Metadata collected for the past three years have informed on the potential drivers of disease severity. The size of the farm and the scale of production, small vs commercial was found to be significant factor in driving disease severity. We will further analyze the data taking into account the cost of pesticides and fertilizer



sprays for managing the production and the disease.

Since our field metadata surveys failed to gain insights into specific management practices used by the growers, we conducted a field trial in 2021 in EV Smith (North Alabama) and Brewton (South Alabama) research stations in the hope that we test different management options and find their influence on the yield. Since these fields regularly see bacterial spot infections, we decided not to inoculate plants in the field to quantify economic impact of different management practices. Unfortunately, Brewton field trial experiment was unsuccessful due to damage by storms in mid-summer. The fields were later infected by southern blight. We could not rate for bacterial leaf spot in Brewton field. In EV Smith, there was no bacterial spot disease recorded in 2021, thus, influence of treatments could not be assessed for management of BLS. Thus, evaluation of current management practices and their relation to the yield remains to be studied.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.

Include approaches taken to ensure robust and unbiased results.

Potnis lab follows Responsible Conduct of Research guidelines established by federal funding agencies, NSF, and USDA. Each of the greenhouse and growth chamber experiments were conducted in at least 5 biological replicates and at least three experimental replicates to ensure reproducibility and replicability. All data generated in this project has been backed up on the BOX folder. For the analysis of sequencing data, we have been using our established and published analysis pipelines (Newberry et al. 2020). Recently, I co-authored article on reproducible research in Phytopathology: [Open Access and Reproducibility in Plant Pathology Research: Guidelines and Best Practices](#) by Niklaus J Grünwald, Clive Bock, Jeff H Chang, Alessandra Alves de Souza, Emerson Del Ponte, Lindsey J du Toit, Anne Elizabeth Dorrance, Jeremiah KS Dung, David H Gent, Erica M Goss, Tiffany Lowe-Power, Laurence V Madden, Frank N Martin, John M McDowell, Rachel Naegele, Neha Potnis, Lina Quesada-Ocampo, George W Sundin, Lindsey Thiessen, Boris Vinatzer, Quan Zeng. T

- 2.8. List key stakeholders that could be served by results of this project.
[Growers, extension agents, and seed industry](#)

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not



designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

During the three-year project period, we have established connections with the growers and extension agents in AL, SC, GA, and NC. We have had a good communication with the extension agents and through them to the growers to keep them updated on the disease analysis of their samples. The extension agents have communicated the concerns of inconsistent weather patterns that are making management decisions difficult for the growers in the past two years. We continue to highlight the importance of clean seeds to the growers as our findings indicate that global pathogen diversity is all circulating in the southeastern US fields. The extension agents have included the findings from our work in their presentations for the growers.

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

A major challenge that we encountered during this project was right when we began our project in early 2020. COVID-related disruptions delayed hiring and some of our planned greenhouse experiments. COVID-related disruptions and restrictions also impeded our field visits. However, we had excellent support from the extension agents and faculty from GA, SC, NC, and AL, who poured in their help as much as they can to visit the fields and collect samples for us.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Weekly meetings are arranged for students and postdocs in Potnis lab. These weekly meetings involve discussion of results, interpretations, and design of the next experiments. Both graduate students and postdocs develop individual development plan at the beginning of their program, and we discuss the self-evaluation of progress at the end of each semester that involves implementation of individual development plan and proposed changes based



on the individual progress. Students and postdocs obtain opportunities to present their work at the conferences and other research symposia organized on campus. These activities are evident from the information product (conference proceedings/abstracts) indicated below. As indicated above, 9 graduate students have been mentored as a part of this project, of which 7 have successfully completed MS or PhD programs.

We held collaboration meetings with other collaborators as we made progress in this project. In addition to the proposed collaborators, one additional collaborator joined this project in the third year of the project, Roberto Molinari, who has helped in building mathematical models to analyze the predictor variables responsible for continued disease outbreaks of BLS.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Following graduate students were funded and mentored through this project.

1. Rishi Bhandari
2. Amanpreet Kaur
3. Ivory Russell
4. Destiny Brokaw
5. Shreya Sadhukhan
6. Pritam Mitra
7. Ranlin Liu
8. Kyle Lesinger
9. Tayler Schillerberg

Following research associate/postdoctoral researchers were funded and mentored during this project.

1. Auston Holland
2. Ashish Adhikari

In addition, one high school student was mentored during the course of this project. The student conducted machine learning analysis on the synthetic microbial community experiments. Mentoring high school student was not proposed in the project. However, since this student indicated interest in applying her skills to the datasets from this project, I mentored her for this project that she presented at the regional science fair competition.



3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Peer-reviewed scholarly Publications
Conference proceedings/abstracts
Invited talks

- 3.2. Please provide a list of citations for the information products produced during the project.

Peer reviewed Publications:

1. Sadhukhan, S., Jacques, M.A., and **Potnis, N.** 2024. Influence of co-occurring weakly pathogenic bacterial species on bacterial spot disease dynamics on tomato. *Plant Disease* 108(1):190-199. Doi: [10.1094/PDIS-05-23-0837-RE](https://doi.org/10.1094/PDIS-05-23-0837-RE)
2. Bhandari, R., Sanz-Saez, A., Leisner, C., and **Potnis, N.** 2023. *Xanthomonas* infection and ozone stress distinctly influence the microbial community structure and interactions in the pepper phyllosphere. *ISME Communications* 3(1), 24. Doi: [10.1038/s43705-023-00232-w](https://doi.org/10.1038/s43705-023-00232-w)
3. Bibi, S., Weis, K., Kaur, A., Bhandari, R., Goss, E., Jones, J.B., and **Potnis, N.** 2023. A brief evaluation of a copper resistance mobile genetic island in the bacterial leaf spot pathogen *Xanthomonas euvesicatoria* pv. *perforans*. *Phytopathology* 113(8): 1394-1398. [10.1094/PHYTO-02-23-0077-SC](https://doi.org/10.1094/PHYTO-02-23-0077-SC)
4. Leisner, C. P., **Potnis, N.**, and Sanz-Saez, A. 2023. Crosstalk and trade-offs: Plant responses to climate change-associated abiotic and biotic stresses. *Plant, Cell & Environment* n/a. doi: [10.1111/pce.14532](https://doi.org/10.1111/pce.14532).
5. **Garrett, K. A.**, D. P. Bebbler, B. A. Etherton, K. M. Gold, A. I. Plex Sulá, and M. G. Selvaraj. 2022. Climate change effects on pathogen emergence: artificial intelligence to translate big data for mitigation. *Annual Review of Phytopathology* 60:357-378.
6. Osdaghi, E., Sharma, A., Goss, E.M., Abrahamian, P., Newberry, E., **Potnis, N.**, Carvalho, R., Choudhary, M., Paret, M., Timilsina, S., Vallad, G., Jones, J.B. 2021. A Centenary for Bacterial Spot of Tomato and Pepper. *Molecular Plant Pathology, Pathogen Profile*. 22(12):1500. <https://bsppjournals.onlinelibrary.wiley.com/doi/epdf/10.1111/mpp.13125>
7. **Potnis, N.** 2021. Harnessing Eco-Evolutionary Dynamics of *Xanthomonads* on Tomato and Pepper to Tackle New Problems of an Old Disease. *Annual Review of Phytopathology*, 59 (1): null. <https://doi.org/10.1146/annurev-phyto-020620-101612>.
8. Stam, R., Gladieux, P., Vinatzer, B. A., Goss, E. M., **Potnis, N.**, Candresse, T., Brewer, M. T. 2021. Population Genomic- and Phylogenomic-Enabled Advances to

- Increase Insight Into Pathogen Biology and Epidemiology. *Phytopathology* 111 (1): 8–11. <https://doi.org/10.1094/PHYTO-11-20-0528-FI>.
9. Shantharaj, D., Williams, M. A., **Potnis, N.**, Liles, M. R. 2021. Burkholderia Gladioli C101 Metabolites Protect Tomato Plants against Xanthomonas Perforans Infection. *J Plant Dis Prot*, 128 (2): 379–390. <https://doi.org/10.1007/s41348-020-00416-9>.
 10. Newberry, E., Bhandari, R., Kemble, J., Sikora, E., and **Potnis, N.** 2020. Genome-Resolved Metagenomics to Study Co-Occurrence Patterns and Intraspecific Heterogeneity among Plant Pathogen Metapopulations. *Environmental Microbiology* 22(7):2693-2708. <https://doi.org/10.1111/1462-2920.14989>.

Conference proceedings/talks:

1. Bhandari, R. and N. Potnis. Biotic and abiotic stress distinctly influence the phyllosphere microbial community structure. Phytobiome Alliance webinar. Sept. 13 2023 (Invited oral presentation).
2. Kaur A., Russell I., Lui R., Bhandari R., Holland A., Potnis N. Rapid adaptation of a pathogen population in response to climatic fluctuation. Annual Meeting of the Southeastern Branch of the American Society for Microbiology, Auburn, AL, November 3rd-5th, 2023. (Oral presentation)
3. Potnis, N. Understanding the drivers of the pathogen population dynamics using strain-resolved metagenomics. Oral presentation in Population Genomics Session, International Congress of Plant Pathology, August 2023. Lyon, France (Oral presentation).
4. Potnis, N. Scanning through a wealth of genomic and metagenomic datasets for the quest of reliable disease diagnostic solutions. Invited presentation at EURL workshop at International Congress of Plant Pathology, August 2023. Lyon, France (Invited Oral presentation).
5. Russell, I., Potnis, N. Differential strain dynamics under host and environmental stressors: A pathway to better understanding pathogen presence and fitness. ASM Microbe 2023. (Oral presentation).
6. Bhandari, R., Potnis, N. Xanthomonas infection and ozone stress distinctly influence the microbial community structure and interactions in the pepper phyllosphere. Plant Health 2023. August 14, 2023, Denver CO (Poster presentation)
7. Bhandari, R., Pena, M, Bowers, R., Weis, K., Newberry, E., Wagner, N., et al. 2023. Genetic and functional diversity to explain generalist and specialist lifestyles in the genus Xanthomonas. Plant Health 2023. August 14, 2023, Denver CO (Poster presentation).
8. Kaur A., Russell I., Potnis N. Analyzing the genomic variations in the pathogenic population under host genotype x environment x pathogen interactions. 12th International Congress of Plant Pathology, Lyon, France, August 21-25th, 2023. (Poster presentation).
9. Weis, K., Potnis, N. Diversity of Mobile Genetic Elements and Associated Fitness Factors to Elucidate the Evolution of Xanthomonas. Genome Biology and Evolution, not yet published, 12th International Congress of Plant Pathology, Lyon, France, August 2023. (Poster presentation).



10. [Potnis, N.](#) Monitoring fine-scale adaptations in pathogen populations in agricultural systems. Xylella files seminar series (online) - February 14, 2023. (Invited Oral presentation)
 11. [Potnis, N.](#), Bennett, R., Broaders, K. The U.S Culture collection network - A central resource for microbial culture collections and their users. USCCN webinar. On Dec 14, 2023 (Invited Oral presentation)
 12. [Bhandari, R.](#), Potnis, N. Resilience of microbial community associated with phyllosphere in response to biotic and abiotic stress. Phyllosphere 2022. 11th International Symposium on Leaf Surface Microbiology, Davis, California, July 2022. (Selected for oral presentation).
 13. Potnis, N. Disease emergence in plant pathogenic bacteria. Fall 2022. Invited seminar speaker, Department of Microbiology and Plant Pathology, UC Riverside.
 14. Potnis, N. Monitoring fine-scale adaptations in pathogen populations in agricultural systems. International Conference of Plant Pathogenic Bacteria ICPPB, Italy. Keynote speaker. July 2022.
 15. Potnis, N. Understanding the basis of host adaptation in a stealthy plant pathogenic bacterium, Xanthomonas. Postdoc Invited seminar speaker, Oregon State University, Department of Botany and Plant Pathology. April 2022.
 16. Potnis, N. Understanding the basis of host adaptation in a stealthy plant pathogenic bacterium, Xanthomonas. Keynote speaker. EuroXanth COST Action 4th Annual Conference. Summer 2021.
 17. Potnis, N. Uncovering the basis of host adaptation in a stealthy plant pathogenic bacterium, Xanthomonas. Department of Plant Pathology, The Ohio State University. Spring 2021 seminar series.
 18. Potnis, N. Revisiting plant disease management in the 'omics era. Biosystems Engineering Department, Auburn University. Spring 2021 seminar series.
 19. Potnis, N. Competition, stress and host defense: How do plant pathogenic bacteria manage them all? Plant Pathology and Plant-Microbe Biology, Cornell University. Graduate student invited seminar speaker Ithaca-Fall 2020.
 20. Potnis, N. Moving beyond binary plant-pathogen interactions for surveillance and management of plant diseases. Plant Pathology and Plant-Microbe Biology, Cornell University. Graduate student invited seminar speaker Geneva-Fall 2020.
 21. Potnis, N. 2020. Understanding infection strategies of an endemic stealthy plant pathogenic bacterium. Fall 2020 Seminar series speaker in School of Plant Sciences, Virginia Tech.
 22. [Bhandari, R.](#), Potnis, N. Tomato disease workshop 2020. (selected oral presentation)
 23. [Potnis, N.](#) Whole genome sequencing of Plant Pathogenic Bacteria. APS Special session on taxonomy. 2020 APS Annual meeting.
- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Nothing to report.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Nothing to report.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s). Nothing to report.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Nothing to report.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

We deposit the raw data as well as scripts used for analyzing the data on our lab GitHub page (potnislabs) and include data availability statements in the published articles. In addition, we built a tomato microbial isolate collection as a part of this project. This collection has been registered at United States Culture Collection Network (USCCN) registry website. The availability of this collection to the



research community was discussed in the webinar presentation and panel discussion organized by USCCN in December 2023.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.
- a) Shotgun metagenomic reads of phyllosphere samples from tomato fields in the southeastern US for 2020, 2021, and 2022 growing seasons.
 - b) Field metadata
 - c) Climatic parameters data
 - d) Amplicon sequencing reads for tomato rhizosphere, endosphere, and phyllosphere samples collected from *Xanthomonas*-infected tomato plants.
 - e) Processed amplicon and shotgun metagenome data for alpha and beta diversity community analyses and community structure and composition analyses.
 - f) Modeling scripts and statistical analysis scripts for predicting risk factors for disease outbreaks
- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

These datasets are related only because they all are related to field survey and contain information on different parameters. The datasets themselves do not contain overlapping data when considered individually.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Metadata files have been made available on the lab GitHub page <https://github.com/Potnislav> for each of the published data. Similar to these published articles, metadata records associated with shotgun metagenomics will be made available alongside the manuscript on the GitHub page and on NCBI BioSample records.



Certification

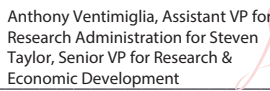
The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Neha Potnis

PI Signature:  Date: 04/02/2024

Authorized Signing Official (ASO) Name: Anthony Ventimiglia, Assistant VP for Research Administration for SVPRED

ASO Signature:  Digitally signed by Anthony Ventimiglia, Assistant VP for Research Administration for Steven Taylor, Senior VP for Research & Economic Development Date: 2024.04.02 09:08:35 -05'00' Date: April 2, 2024